

Engineer who owns the product: from ambiguous brief to shipped system — across AI agents, real-time graphics, and AR/VR. Built the agentic AI architecture behind **Create with AI**, TikTok Effect House's AI generation tool that turns effect ideas into real AR effects, live on TikTok. Client-facing agency background: IBM, Hershey, Santander (Webby Award). U.S. work authorization (no sponsorship).

## EXPERIENCE

### Senior Research Engineer (AI Product) · ByteDance / TikTok Jan 2024 – Present

- **Architected AIGE**, the agentic AI system behind "Create with AI" — type an effect idea, get a runnable AR effect live on TikTok. Own the end-to-end agent architecture: an LLM classifies, decomposes, and plans against validated schemas; deterministic code executes — across body tracking, voice recognition, and game mechanics.
- Designed AIGE's knowledge & skill retrieval (RAG) — hierarchical knowledge (sceneObject → component → property) grounding MCP tool calls into Effect House; retrievable skill packs per game / effect genre.
- Built AIGE's evaluation system — designed the benchmark and automated validation of generated effects, raising end-to-end generation success measurably release over release.

### Interactive Engineer · ByteDance / TikTok Effect House Jan 2021 – Jan 2024

- Designed & shipped the **Material Graph node system** of the Effect House creation tool — defined the node taxonomy (Transition / 2D SDF / Effect), specced inputs & outputs, and built nodes from prototype to production.
- Led the Template production team — shipped **120+ templates**, driving over **40% of all TikTok effect creations**.
- Led Effect House community education — 50+ tutorial videos and regular YouTube live sessions; scaled to **400k+ Discord members** and **64k YouTube subscribers**.

### Creative Technologist · Havas Worldwide, New York Jun 2017 – Dec 2020

- Shipped AR/VR/WebGL products across 4 years of client-facing agency work — translating ambiguous briefs from IBM, Hershey, VUSE, Santander, Adidas, and Hugo Boss into production, owning storyboards, architecture, and production code.
- **Webby Award winner** (2019, Augmented Reality) — "In Someone Else's Shoes" for Santander.
- Lead creative technologist on IBM Watson Holobot (IBM Think 2018) and Hershey's Pop Kiss Studio VR retail installation — interaction systems, particle VFX, and on-site user testing with real customers.

### Adjunct Faculty · Parsons School of Design Jan – May 2019

- Taught "Warping Reality" — AR development with Unity, ARKit/ARCore, Vuforia.

## SELECTED PROJECT

### Sound Visualization — Agentic VFX System for Live Music solo · 2026

- An agentic system that turns a song into a live, editable visual performance: three agent stages with typed, human-editable artifacts (music analysis → visual plan → performance score); WebGL runtime with live mic modulation, built for stage projection.

## SKILLS

### AI systems

Agentic architecture & multi-agent pipelines, LLM planning, function calling & MCP, structured outputs, RAG & skill libraries, LLM/VLM evals & benchmarks, AI-assisted dev (Claude Code, Cursor)

### Engineering

JavaScript/TypeScript, GLSL / WebGL / Three.js, C# / Unity3D, Python, React, Node, audio DSP

### Interaction & visual craft

Interaction design, motion design, real-time visual aesthetics (shaders, particles), Figma

## EDUCATION

### New York University

M.P.S., Interactive Telecommunications Program (ITP)  
Red Burns Scholarship · 2015 – 2017

### Southeast University

B.Eng., Biomedical Engineering  
Outstanding Graduate · 2009 – 2013